# Filtrations through a finite set of formulas are finite

The Open Logic Project contributors

openlogic:normal-modal-logic/filtrations:local-snapshot-20261013;
CC BY 4.0

*Editorial scope.* Statement and complete author proof are included verbatim as the span lines 1091-1112 of the pinned original source; no text is rewritten or paraphrased.

*Allowed standard prerequisites (editorial).*

- filtration of a modal model through a set of formulas Gamma
- the equivalence relation of agreeing on all formulas in Gamma
- finite powerset cardinality

**Proposition 0.1.** *If* $\Gamma$ *is finite then any filtration* $M^* \models$ *of a model* $M \models$ *through* $\Gamma$ *is also finite.*

*Proof.* The size of $W^*$ is the number of different classes $[w]$ under the equivalence relation $\equiv$. Any two worlds $u$, $v$ in such class—that is, any $u$ and $v$ such that $u \equiv v$—agree on all !!formulas $!A$ in $\Gamma$, $!A \in \Gamma$ either $!A$ is true at both $u$ and $v$, or at neither. So each class $[w]$ corresponds to subset of $\Gamma$, namely the set of all $!A \in \Gamma$ such that $!A$ is true at the worlds in $[w]$. No two different classes $[u]$ and $[v]$ correspond to the same subset of $\Gamma$. For if the set of !!formulas true at $u$ and that of !!formulas true at $v$ are the same, then $u$ and $v$ agree on all formulas in $\Gamma$, i.e., $u \equiv v$. But then $[u] = [v]$. So, there is !!ainjective function from $W^*$ to $\mathcal{P}\Gamma$, and hence $|W^*| \leq |\mathcal{P}\Gamma|$. Hence if $\Gamma$ contains $n$ sentences, the cardinality of $W^*$ is no greater than $2^n$. □

*Proof.* The size of $W^*$ is the number of different classes $[w]$ under the equivalence relation $\equiv$. Any two worlds $u$, $v$ in such class—that is, any $u$ and $v$ such that $u \equiv v$—agree on all !!formulas $!A$ in $\Gamma$, $!A \in \Gamma$ either $!A$ is true at both $u$ and $v$, or at neither. So each class $[w]$ corresponds to subset of $\Gamma$, namely the set of all $!A \in \Gamma$ such that $!A$ is true at the worlds in $[w]$. No two different classes $[u]$ and $[v]$ correspond to the same subset of $\Gamma$. For if the set of !!formulas true at $u$ and that of !!formulas true at $v$ are the same, then $u$ and $v$ agree on all formulas in $\Gamma$, i.e., $u \equiv v$. But then $[u] = [v]$. So, there is !!ainjective function from $W^*$ to $\mathcal{P}\Gamma$, and hence $|W^*| \leq |\mathcal{P}\Gamma|$. Hence if $\Gamma$ contains $n$ sentences, the cardinality of $W^*$ is no greater than $2^n$. □